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Methods for estimating energy demand for electric refrigerated vehicles

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Abstract

The development of electromobility in the transport sector is gaining importance due to increasingly strict emission regulations and the need to reduce CO_2 emissions. Electric refrigerated vehicles (EVRs) present a promising alternative to traditional refrigerated transport, combining zero-emission propulsion with temperature control systems. However, estimating the energy demand of EVRs is a complex task, as it must account for both vehicle propulsion and refrigeration under dynamic operating conditions. This paper reviews current literature, discusses various estimation methods — including analytical, measurement-based, simulation, and normative approaches — and presents a modular concept for a comprehensive energy demand estimation method. The proposed framework integrates technical vehicle parameters, environmental conditions, and transport scenarios to provide accurate, simulation-based predictions. Its application can support the design and optimization of EVRs in terms of energy efficiency, logistics planning, and battery system sizing.

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Keywords: electric refrigerated vehicles; energy demand; cold chain; simulation model; energy efficiency

1. Introduction

With the growth of environmental awareness and tightening of emission standards, the transport sector is undergoing intensive technological and regulatory transformation, in which the development of electromobility plays a key role as an alternative to traditional combustion engines. Electric refrigerated vehicles are an innovative

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solution that not only reduces CO_2 emissions and noise in cities, but can also bring economic and operational benefits to logistics companies. Every year, refrigerated transport handles 40 to 50 million tons of food, of which almost half are fresh fruit and vegetables [1].

Unlike standard electric vehicles, which are mainly powered by electric engines, refrigerated vehicles (EVR) must meet rigorous performance standards to maintain a constant temperature of the transported goods while preventing sudden temperature changes. Internationally, the most commonly used means of refrigerated transport for transporting food is a refrigerated truck marked with the FRC symbol - i.e. a type F vehicle with reinforced thermal insulation (R) in temperature class C. This type of semi-trailer allows the transport of goods at a controlled temperature, ranging from -20°C to +12°C [2].

The general principles of food transport are given in the "Agreement on the International Carriage of Perishable Foodstuffs and on the Special Equipment to be Used for Such Carriage (ATP)" [2]. One of the key operational challenges is the proper estimation of energy demand not only for the drive, but also for temperature maintenance systems - especially in changing weather conditions and when the cargo space is frequently opened. An important element in the context of energy demand is also the integration of renewable energy sources, such as photovoltaic panels, which can support the power supply of refrigeration units, allowing vehicles to have a longer autonomy and lower energy consumption from batteries.

2. Literature review

The analysis of the literature on the energy demand of electric refrigerated vehicles shows that there are more and more studies examining both technical and economic issues related to their use. It has been shown that environmental factors [3] significantly affect energy consumption, with the key role being played by ambient temperature due to its impact on battery efficiency [4] and the intensity of use of auxiliary devices [5]. Attention is also paid to other external conditions, such as headwind [6], traffic intensity [7] or road surface condition. In addition, some studies indicate that factors related to the driver, including driving style [8], charging method and time [9], work organization and route planning [10], also have a significant impact on the level of energy consumption.

Maiorino et al. [11] created a detailed energy model for a hybrid refrigerated vehicle equipped with photovoltaic panels and batteries to estimate fuel consumption, operating costs and CO_2 emissions. The analyses carried out showed that the use of such a solution can reduce CO_2 emissions by up to 20% compared to traditional refrigerated vehicles. [12] presented an integrated model of an electric refrigerated vehicle, combining TRNSYS and MATLAB computer simulation tools. The study used a genetic algorithm to optimize air flow and refrigerant, aimed at reducing energy consumption. Sharma and Alleyne [13] developed an extreme control method for a battery-powered compressor system. The applied solution allowed for reducing energy consumption and extending battery life by up to 33%. The results of their research confirmed the effectiveness of adaptive control algorithms under variable thermal load conditions. Poks et al. [14] analyzed the potential of using electric vehicles in the distribution of products requiring refrigeration. As part of their research, they developed a model of an electric truck equipped with an electric refrigeration unit, taking into account both the power consumption of the drive system and the energy consumption of the cooling system in various operating conditions. [15] proposed innovative solutions for stabilizing and controlling food temperature fluctuations in refrigerated vehicles, using phase change materials (PCM). The conducted research showed that the selection of appropriate PCM for the external walls of the container is a major challenge due to dynamically changing environmental conditions. Miri I. et al. [16] proposed an analytical model for estimating energy consumption, taking into account variables such as vehicle speed, driving time, road conditions, vehicle weight and technical parameters of the drive system. The results were compared with the actual energy consumption of the electric vehicle. The model showed high accuracy, with an average error of less than 10%. Castañeda et al. [17] proposed a new route optimization model for refrigerated electric vehicles (EVs) taking into account temperature variation over time and its impact on energy consumption for cooling. They prove that the previous models neglecting temperature changes during the day lead to incorrect energy consumption estimates and inefficient route planning. Cooling accounts for as much as 44% of the total energy consumption of a refrigerated EV. The authors indicate the need to expand the charging infrastructure for refrigerated electric vehicles. The authors [18] proposed a quantitative approach to map energy flows throughout the cold chain in the food sector and to assess the overall energy efficiency. The analysis of the results shows that cooling and freezing of goods is not sufficient, because at this stage 43% of quality degradation occurs, and electricity is the main energy carrier (up to

71%).

In order to achieve climate neutrality by 2050 [19], the European Commission is taking numerous actions, one of which is the Regulation 2023/851 adopted by the European Parliament in 2023. It provides for a total reduction of CO2 emissions for new passenger cars and light commercial vehicles starting from 2035 [20].

3. Methods for estimating energy requirements for refrigerated vehicles

The rapid development of electromobility and increasingly stringent standards for reducing greenhouse gas emissions are promoting the popularization of electric refrigerated vehicles used in the transport of goods requiring controlled temperature. Unlike conventional refrigeration systems powered by fuel from independent generators, electric vehicles use energy stored in traction batteries or dedicated energy storage units. Such a system requires a precise estimate of the energy demand for both the vehicle drive and the cooling system, which draw power from the same source. The factors influencing the total energy consumption in the operating cycle are shown in figure 1.

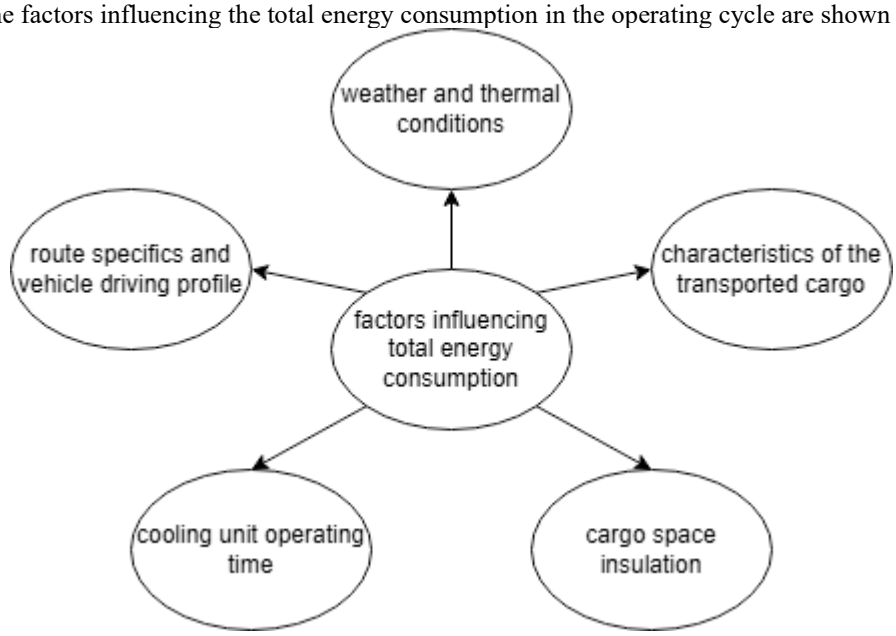

Fig. 1 Factors influencing the total energy consumption in the operating cycle

The methods of estimating the energy demand for refrigerated vehicles can be divided according to their application.

We distinguish:

- Analytical methods (estimation of heat losses and necessary cooling power)
- Measurement methods (actual measurement of energy consumption by the refrigeration unit during vehicle operation in various operating conditions)
- Computer simulation (using specialized software: TRNSYS, MATLAB, CFD)
- Normative methods (legal acts, standards, national guidelines).

The analytical method was used in their research by Chen.Y et al., who used phase change materials (PCM) in refrigerated vehicles as a way to save energy and reduce emissions. PCMs can store and release heat, which allows for stabilization of the temperature inside the cargo space without continuous operation of the refrigeration unit [21]. Hwang et al. focused their research on the analysis of air flow and heat transfer inside the cargo space of a refrigerated truck, identifying factors influencing cooling efficiency [22]. The authors of [11],[23] and [24] used computer simulation methods using specialized software. Giliberto et al. developed a model for optimizing the

cooling strategy in refrigerated vehicles, which takes into account variable operating conditions, such as random door opening times or different initial temperatures of transported products. The proposal combines advanced thermodynamic modeling with multi-stage stochastic programming, which allows for reducing fuel consumption by up to 40% compared to traditional, deterministic cooling management methods [23]. Kang Z. et al. [24] used numerical simulations to analyze heat losses in the cargo space of a medium-sized refrigerated truck, taking into account different configurations and operating conditions.

Energy demand can also be estimated using measurement methods, which consist in the actual measurement of energy consumption by the refrigeration unit during vehicle operation in various operating conditions. Studies using this method were conducted by [25], [26], [17] and [18]. The authors of the study [25] analyzed data from 147 long-distance routes, covering 200,000 km and 84,588 liters of fuel used. Based on this data, they developed a formula estimating the total fuel consumption and CO_2 emissions, taking into account the type of trailer, route, load weight and operating time of the refrigeration unit. The model was characterized by an average error of 6.7%. Wang K. and Du N. [26] published a study on the energy consumption management strategy in refrigerated logistics using the Internet of Things (IoT) technology. The study was conducted under the conditions of a simulated cold chain logistics scenario. The obtained results indicate that, compared to the conventional logistics system, the new system significantly increases energy efficiency, reduces energy consumption by about 20%, increases the precision of temperature and humidity control to 94%, and improves the transport process, shortening its time by 8.33%.

4. Application of new technologies on the market

In the face of growing pressure to achieve climate neutrality, the transport and logistics sector is stepping up its efforts to reduce CO_2 emissions. One key direction is the development of electric refrigerated trailers powered by renewable energy sources, which enable more environmentally friendly transport of temperature-sensitive cargo.

In cooperation with the trailer manufacturer Kögel and the refrigeration unit supplier Carrier, Sono Motors presented at the IAA an innovative refrigerated trailer equipped with 26 polymer photovoltaic panels with a total surface area of 26 m² and a peak power of 4.68 kW. In the typical Munich solar conditions, this system generates an average of 11.8 kWh of energy per day, which is stored in batteries and used to power the Vector HE19 [27] electric refrigeration unit.

The Dutch manufacturer Kraker Trailers has developed a prototype of a moving floor trailer that uses an electric axle to recover energy during braking. The stored energy is stored in 42 kWh batteries and can be used to assist the drive during acceleration or to power a hydraulic pump. This solution allows for fuel savings of 10–20% and a reduction in CO_2 emissions [28].

Schmitz Cargobull has launched the EcoGeneration series of trailers, including the EcoFIX, EcoFLEX and EcoVARIOS models, which feature optimised aerodynamics. Thanks to their adjustable roof height, these trailers can adapt to different loads, reducing air resistance and fuel consumption by up to 5%. In practice, when used correctly, fuel savings can exceed 10%, which translates into a significant reduction in CO_2 emissions. In addition, the EcoFLEX and EcoVARIOS trailers offer flexibility in adjusting the body height, allowing for the efficient transport of different loads while maintaining aerodynamic benefits [29]

5. The concept of the method for estimating energy demand

The aim of this chapter is to present the concept of a method that allows for estimating the energy demand of an electric refrigerated vehicle depending on its design parameters, operating conditions and environmental factors. The proposed method is modular and can be the basis for building a simulation analytical tool.

In principle, the method should take into account two main components of energy consumption:

- The energy required to drive the vehicle, which depends on the vehicle mass, rolling resistance, aerodynamic drag and route profile;
- The energy required to maintain the internal temperature of the cargo compartment, resulting from the thermal balance between the vehicle interior and the external environment.

The method assumes the use of known input data (catalog or measurement) and transforming them into an estimate of energy consumption using simplified physical models and numerical simulations.

5.1 Modules description

Module A – Vehicle & cargo box parameters

Contains data describing the vehicle and cargo box:
- L – cargo box length [m]
- W – cargo box width [m]
- H – cargo box height [m]
- d_{ins} – wall insulation thickness [m]
- λ – thermal conductivity of insulation material $[W/m \cdot K]$
- A – surface area of the cargo box walls $[m^2]$
- α – vehicle body color absorptivity
- m_{veh} - vehicle mass [kg]
- A_f – frontal area of the vehicle $[m^2]$
- C_d – aerodynamic drag coefficient
- U – thermal transmittance coefficient $[W/m^2 \cdot K]$
- V_{cargo} – volume of cargo box $[m^3]$
- C_{bat} – battery capacity [kWh]

Module B – Driving cycle profile

Contains driving dynamics data:
- $v(t)$ – velocity velocity over time [m/s]
- $a(t)$ – vehicle acceleration over time $[m/s^2]$
- $\theta(t)$ – road slope angle [°]
- f_{stop} – stop frequency [1/h]
- l_{route} – route lenght [km]
- C_{roll} – surface type resistance coefficient
- β_{drv} - driver behaviour coefficient
-

Module C – Enviromental conditions

Allows to take into account the variability of external conditions:
- $T_{amb}(t)$ – ambient temperature over time [°C]
- $G(t)$ – solar irradiance over time [W/m^2]
- v_{wind} – wind speed [m/s]
- ϕ – relative humidity [%]
- t – time of the day [h]
- φ – geographic latitude

Module D1 – Thermal load estimation

Estimation of the total heat load of the cooling room (cold chamber), i.e. balancing all heat losses and gains affecting the temperature inside it.
- $\boldsymbol{Q_{loss}}$ $\rightarrow$ ***heat loss*** (***heat transfer to the interior of the chamber***)

$$Q_{loss} = \sum (U_i \cdot A_i \cdot \Delta T) \tag{1}$$

where:
$\boldsymbol{U_i}$ **-** heat transfer coefficient for i-th partition [W/m²K]
$\boldsymbol{A_i}$ **-** area of i-th partition [m²]
$\Delta\boldsymbol{T}$ **-** temperature difference between the environment and the interior of the chamber [K]

Additionally, heat loss when opening the door can be taken into account:

$$Q = n \cdot V \cdot \rho \cdot C_p \cdot \Delta T \tag{2}$$

where:
$\boldsymbol{n}$- number of air changes per hour
$\boldsymbol{V}$- chamber volume [m³]
$\boldsymbol{\rho}$ - air density [kg/m³]
$\boldsymbol{C_p}$ - specific heat of air [kJ/kgK]
$\Delta\boldsymbol{T}$ - temperature difference between the environment and the interior of the chamber [K]

Heat generated inside the chamber or introduced with the goods.

- $\boldsymbol{Q_{gain}} \rightarrow$ ***heat gains*** (***heat source inside the chamber***)

1) Heat released by products during cooling

$$Q = m \cdot C_w \cdot \Delta T \tag{3}$$

where:
$\boldsymbol{m}$- product weight [kg]
$\boldsymbol{C_w}$- specific heat of the product [kJ/kgK]
$\Delta\boldsymbol{T}$ - the difference in product temperature from the introduction until reaching the temperature in the chamber [K]

2) Heat from people and equipment working in the chamber

$$Q = N \cdot q_L$$
$$Q = N_u \cdot q_u \tag{4}$$

where:
$\boldsymbol{N}$- number of people
$\boldsymbol{q_L}$- specific heat emitted by a person [W]
$\boldsymbol{N_u}$- number of devices
$\boldsymbol{q_u}$- specific heat emitted by the device [W]

3) Heat from lighting in the chamber

$$Q_o = P_o \cdot \tau \tag{5}$$

where:
$\boldsymbol{P_o}$- lighting power
$\boldsymbol{\tau}$ – working time

Heat balance in the static and moving state

Heat balance in the static state – includes only heat losses through partitions and gains from permanent sources (goods, devices, lighting operating in standby mode):
-constant, relatively low level of heat load.
-no air exchanges through door opening.
Heat balance in motion (chamber operation) – additionally:
-losses through infiltration when opening doors.
-gains from new goods brought in at a higher temperature.
-heat from people and operating devices.
-higher momentary heat load.

Module D2 – Refrigeration System Model

This module is an extension of the classic heat balance with an energy model of the cooling system. Its main task is to convert the instantaneous heat flux that must be collected from the cooling chamber into the actual electricity consumption. The basic formulas describing the operation of this module are:

- Electrical energy consumed by the refrigeration unit, which is based on the general definition of the COP and is commonly found in the literature on refrigeration systems [30]:

$$E_{cool} = \sum_{i=1}^{n} \left(\frac{Q_{total}\,(t_i)}{COP(t_i)} \cdot \Delta t \right) \tag{6}$$

where:
E_{cool} – total electrical energy consumed by the cooling unit over time [Wh]
$Q_{total}\,(t_i)$ - instantaneous heat gain to the cargo space at time step t_i [W]
$COP(t_i)$ - Coefficient of Performance of the cooling system at time step t_i
Δt – time step interval [h]
n – total number of time steps in the simulation

- Time-varying Coefficient of Performance [31]:

$$COP = \frac{Q_{cool}}{P_{el}} \tag{7}$$

where:
Q_{cool} – supplied cooling power [W]
P_{el} - electrical power consumed by the system [W]

Module E – Total Energy Demand Calculation

$$E_{total} = E_{drive} + E_{cool} \tag{8}$$

where:
E_{total} - total energy required for the complete delivery cycle [Wh]
E_{drive} – energy required to propel the vehicle [Wh]
E_{cool} – Energy consumed by the refigeration system [Wh]

5.2 System description

Vehicle and cargo space parameters (module A) affect how much heat can penetrate the vehicle interior – so they are the basis for estimating heat losses. Environmental conditions (module C) directly influence the intensity of these heat losses – the warmer the environment, the greater the cooling load. Based on the total heat load calculated in module D1, module D2 determines how much energy the cooling system must use to maintain the set temperature inside the vehicle. The result of the D2 module operation is the energy used for cooling (E_cool) depending on the system efficiency (e.g. COP). Separately, the driving cycle profile (module B) influences the energy consumption of the drive (E_drive) - depending on weight, speed and route shape. Finally, both sources of energy consumption (propulsion and cooling) are added together in Module E, giving the final result: the total energy requirement of the vehicle. The proposed model is shown in figure 2.

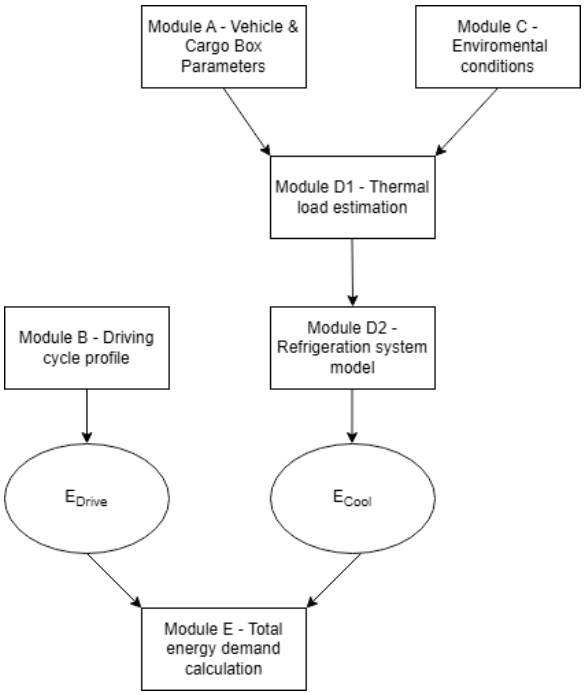

Fig. 2 Block diagram of the conceptual model for estimating energy demand

5.3 Possibilities of applying the concept

The proposed method can be used in analyses of:

- comparative analyses for different variants of vehicles and refrigeration units,
- assessment of the impact of insulation and logistics strategies on energy consumption,
- designing power supply and selecting battery capacity,
- optimization of routes to minimize energy losses.

Conclusion

Estimating the energy demand of electric refrigerated vehicles is a multidimensional challenge that requires a holistic approach combining technical, environmental, and operational factors. Unlike conventional vehicles, EVRs must supply energy not only for propulsion but also for maintaining a stable thermal environment within the cargo compartment. This dual demand places unique constraints on battery capacity, system design, and route planning. As shown in the literature, the accuracy of energy consumption estimations can vary significantly depending on the method used, with each approach — whether analytical, measurement-based, simulation-driven, or normative — offering specific advantages and limitations.

The modular method proposed in this study provides a flexible and scalable framework for estimating energy demand under varying real-world conditions. By dividing the estimation process into dedicated modules that account for vehicle characteristics, driving cycles, environmental conditions, thermal loads, and refrigeration system efficiency, it becomes possible to simulate complex interactions and predict total energy consumption with greater precision. Such an approach not only aids in improving the design of energy systems, including battery capacity and

thermal insulation, but also supports the development of optimized logistics strategies that reduce operational costs and carbon emissions.

Furthermore, the proposed model can serve as a foundation for further development of advanced simulation tools and digital twins for EVRs, enabling ongoing performance monitoring and predictive energy management. As the transportation sector continues to decarbonize, tools that enable accurate, data-driven decisions about vehicle design and fleet operation will play a crucial role. By integrating renewable energy sources, such as solar panels, and adopting smart control strategies, the energy efficiency and autonomy of EVRs can be significantly enhanced.

Ultimately, the ability to reliably estimate and optimize the energy consumption of refrigerated electric vehicles is not only essential for their operational viability but also for achieving broader environmental goals. The work presented in this paper contributes to that aim by offering both a synthesis of current knowledge and a practical, adaptable method for energy modeling in the evolving landscape of sustainable transport.

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